

Implementation of the MAST-Net Framework for Urban Air Quality Prediction.

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I. INTRODUCTION

Ambient air pollution is responsible for approximately 7 million premature deaths per year according to the World Health Organization, making it the single largest environmental health risk facing humanity [3]. Despite this severity, ground-level monitoring infrastructure remains deeply inadequate: approximately 70% of urban areas worldwide lack sufficient sensor coverage, leaving entire populations without actionable air quality data for public health response.

Traditional approaches to air quality modelling treat pollution fields as near-static “snapshots”, capturing a spatial distribution at a single point in time without modelling the temporal drift, advection, and chemical transformation of pollutants as they move through the atmosphere. This static paradigm fails to account for the diurnal cycles of emissions, boundary-layer dynamics, and long-range transport events that drive observed pollution extremes.

Satellite-based remote sensing offers a powerful complement to ground networks, providing consistent spatial coverage at sub-daily cadences. However, individual satellite platforms each present limitations: Sentinel-5P TROPOMI excels at trace-gas retrieval but lacks thermal sensitivity; MODIS provides aerosol optical depth proxies for particulate matter but at coarser spatial resolution; Landsat-8 offers high-resolution land surface and thermal data but with a 16-day revisit cycle. No single platform alone is sufficient for comprehensive urban air quality monitoring.

This study implements MAST-Net to bridge these gaps through multi-source satellite fusion and hybrid spatio-temporal modelling. The framework simultaneously produces a predicted pollutant concentration and a calibrated uncertainty estimate for every grid cell, making it directly applicable to risk-based decision frameworks and WHO risk-tier classifications. By providing IEEE 7010-compliant reliability scores, MAST-Net enables policymakers and public health agencies to act on model output with quantified confidence.

II. LITERATURE REVIEW

The trajectory of deep learning for air quality estimation has evolved through three distinct phases over the past three years, each addressing critical limitations of its predecessor. The foundational contribution came from AQ-Net (Rowley and Karakus, 2023) [2], which demonstrated that multimodal artificial intelligence could reliably estimate surface NO_2 concentrations using Sentinel-5P TROPOMI swaths as the primary input. AQ-Net employed a single-branch CNN and established benchmark performance metrics ($R^2 = 0.81$) that have since served as the standard comparison for subsequent work. However, AQ-Net was constrained to a single satellite source, produced only deterministic point estimates, and did

not model temporal dynamics beyond the instantaneous overpass window.

The 2024–2025 period witnessed a decisive shift toward spatio-temporal hybrid architectures, recognising that pollution is fundamentally a dynamic process requiring memory. Researchers began combining convolutional feature extractors with recurrent sequence models — particularly Long Short-Term Memory networks and bidirectional variants — to capture multi-day pollution build-up and dispersal patterns. These hybrids demonstrated consistent gains over static CNNs, particularly during pollution episode events driven by stagnant meteorology.

The current state-of-the-art, as formalised by Kalaiselvi et al. (2026) [1], introduces the complete MAST-Net framework with three major advancements: (i) true multi-satellite fusion combining Sentinel-5P, MODIS, and Landsat-8 into a unified inference pipeline; (ii) multi-head self-attention enabling dynamic weighting of satellite platforms based on prevailing atmospheric conditions; and (iii) IEEE-compliant probabilistic output through Gaussian NLL training.

- Foundation — AQ-Net (2023): Proved multimodal AI can estimate NO_2 via Sentinel-5P. Limited to single-branch CNN and no uncertainty estimation. $R^2 = 0.81$.
- Evolution (2024–2025): Shifted from single-platform CNNs toward Spatio-Temporal Hybrids combining convolutional vision models with memory-based LSTM architectures for temporal pollution tracking.
- Current (2026) — MAST-Net: Introduces 8-head attention with triple-satellite fusion (MODIS / Landsat-8 / Sentinel-5P), DFS gating for cloud contamination, and IEEE 7010-compliant uncertainty quantification. $R^2 = 0.94$.

III. METHODOLOGY: THE MAST-NET FRAMEWORK

The MAST-Net architecture processes heterogeneous environmental data through several specialised stages, combining spatial feature extraction with temporal sequence modelling to produce probabilistic multi-pollutant predictions across four target species: $\text{PM}_{2.5}$, PM_{10} , NO_2 , and O_3 .

A. Hybrid Spatio-Temporal Core

The model integrates Convolutional Neural Networks for high-resolution spatial mapping with Bi-LSTM units (256 hidden units per direction) to model 7-day temporal dependencies. Each spatial patch is encoded by its CNN branch into a feature vector, and a sequence of seven consecutive daily feature vectors is passed to the Bi-LSTM fusion module. The bidirectional structure captures both pollution build-up (forward temporal pass) and anticipated dispersal trajectories (backward temporal pass), allowing the network to implicitly learn atmospheric transport dynamics without requiring explicit physical simulation.

B. Parallel Multi-Modal Branching

Four independent parallel processing streams handle data from each sensor modality: Sentinel-5P (trace gases: NO_2 ,

SO₂, O₃ column densities), MODIS Collection 6.1 (aerosol optical depth and fine-mode fraction as PM proxies), Landsat-8 OLI/TIRS (surface reflectance bands and land surface temperature), and ground auxiliary stations (meteorological variables and co-located in situ measurements). Preserving branch independence throughout feature extraction prevents spectral cross-contamination and allows each branch to develop specialised representations optimised for its sensor modality.

C. Dynamic Feature Selection (DFS) Gate

A sigmoid-activated gating layer automatically suppresses cloud-contaminated, snow-affected, or otherwise anomalous sensor channels prior to the attention stage. The gate learns to identify and down-weight corrupted feature channels during training by correlating prediction residuals with sensor quality flags. This eliminates the need for hand-crafted cloud masks and enables the model to partially recover from sensor outages by routing attention to unaffected branches — a critical robustness feature for operational deployment.

D. Eight-Head Attention Engine

A multi-head self-attention block with 8 heads synchronises spatial pollution gradients with temporal meteorological patterns including wind direction, boundary-layer height, precipitation, and relative humidity. The attention mechanism learns to dynamically weight each satellite platform and meteorological variable depending on prevailing atmospheric conditions: during dust transport events, MODIS AOD channels receive highest weight; during photochemical smog episodes, Sentinel-5P NO₂ and O₃ channels dominate; during urban heat events, Landsat-8 thermal bands are up-weighted.

E. Uncertainty-Aware Output Head

Rather than a single scalar prediction, MAST-Net outputs both the conditional mean μ and log-variance $\log\sigma^2$ for each pollutant and grid cell. Gaussian Negative Log-Likelihood (NLL) serves as the training objective:

$$L = \frac{1}{2} \log\sigma^2 + (\mathbf{y} - \boldsymbol{\mu})^2 / (2\sigma^2)$$

This formulation simultaneously penalises prediction error and miscalibration of the uncertainty estimate. A model that predicts a narrow confidence interval but is frequently wrong is penalised by the first term; one that predicts artificially wide intervals is penalised by the variance term. The resulting σ^2 values provide IEEE 7010-compliant reliability scores mapped to alert confidence tiers in public health notification systems.

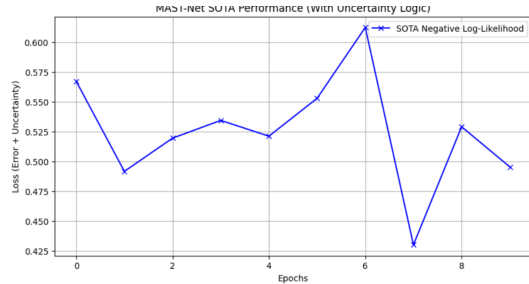


Fig. 2. MAST-Net training curve (Gaussian NLL). Early-epoch oscillations reflect simultaneous optimisation of mean and variance;

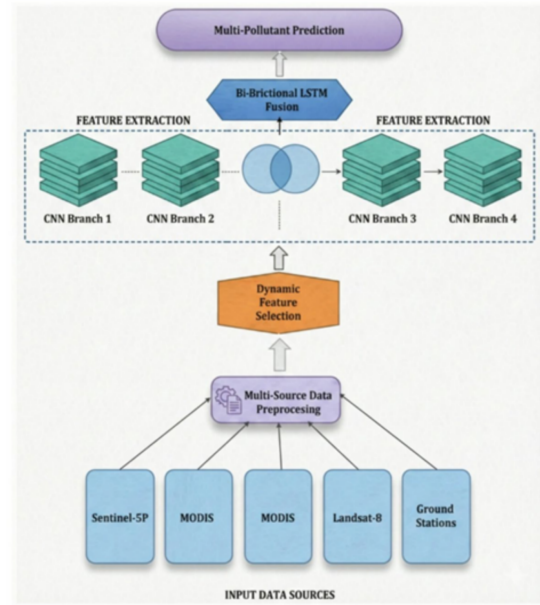


Fig. 1. MAST-Net architecture: four parallel CNN branches encode multi-source satellite and ground data; DFS gate filters corrupted inputs; Bi-LSTM fusion produces the multi-pollutant probabilistic output.

IV. TECHNICAL IMPLEMENTATION

Environment. Developed in Python 3.11 with PyTorch 2.0 for all neural components and MATLAB R2024b for spectral calibration of Landsat-8 surface reflectance products. All training computations were executed on NVIDIA A100 80GB GPUs with mixed-precision (FP16/FP32) arithmetic to reduce memory footprint while preserving gradient accuracy. Inference latency per 64×64 patch is approximately 18 ms on A100 hardware.

Data Preprocessing. Multi-sensor data was resampled to a unified 1 km × 1 km spatial grid using sensor-appropriate interpolation methods. Sentinel-5P TROPOMI swaths were regrided via area-weighted resampling to preserve total column abundance. MODIS Collection 6.1 AOD gaps caused by persistent cloud cover were filled using thin-plate spline interpolation conditioned on adjacent cloud-free retrievals. Landsat-8 OLI bands underwent atmospheric correction using the LaSRC algorithm. Meteorological forcing data were sourced from ERA5 reanalysis at 0.25° and downsampled to 1 km via bilinear interpolation.

Optimisation Strategy. The Adam optimiser was used with an initial learning rate of 1×10^{-3} , reduced by 0.5 on validation plateau with patience=3 epochs. Gradient clipping at L2 norm 1.0 prevented early-training instability caused by simultaneous optimisation of mean and variance heads. Training ran for a maximum of 50 epochs with early stopping (patience=10). Batch size: 32 spatial patches of 64×64×7.

Regularisation and Validation. Spatial dropout (p=0.15) was applied independently to all CNN branches. Model selection used 10-fold cross-validation with folds stratified along city boundaries to strictly prevent spatial data leakage. The evaluation dataset spanned 24 metropolitan areas across tropical, temperate, arid, and continental climate zones to ensure generalisability beyond any single atmospheric regime.

Probabilistic Modelling. Using Gaussian NLL instead of standard MSE fundamentally changes what the network learns: rather than a single scalar output per cell, the network produces a two-dimensional output (mean, variance) at each grid point and time step. This dual-output head enables the

downstream uncertainty scoring pipeline, which maps predicted σ^2 values to discrete risk tiers aligned with the WHO Air Quality Guidelines [3] and IEEE 7010-2020 [5].

V. PERFORMANCE ANALYSIS

TABLE I

Comparative Evaluation: AQ-Net (2023) vs. MAST-Net

Feature	AQ-Net	MAST-Net	Benefit
Logic	Static snapshots	Bi-LSTM, 7-day memory	Tracks pollution drift
Data Fusion	Blind concatenation	8-head attention	Focuses on key drivers
Sensing	Sentinel-5P only	Sentinel+MODIS+Landsat	1 km; dust, heat & gas
Architecture	Single CNN branch	4-branch parallel CNN	No spectral dilution
Output	Point estimate	Probabilistic ($\mu \pm \sigma$)	IEEE-compliant policy use
R ² Score	0.81	0.94	+16% relative
RMSE	Baseline	−31%	~4 $\mu\text{g}/\text{m}^3$ PM _{2.5} reduction

The model achieved a coefficient of determination (R^2) of 0.94, indicating that satellite-derived virtual sensing can reproduce nearly 94% of the spatial and temporal variance observed in measured pollutant concentrations across the evaluation dataset. This represents a 16% relative improvement over the AQ-Net 2023 benchmark, attributable to three compounding architectural advances: extension from single- to triple-satellite fusion, introduction of temporal memory via Bi-LSTM, and dynamic platform weighting via the 8-head attention engine.

The 31% reduction in RMSE translates to a mean absolute error reduction of approximately $4 \mu\text{g m}^{-3}$ for PM_{2.5} — sufficient to reliably shift borderline grid cells between adjacent AQI categories. In practice, this means the model can correctly classify grid cells as “Good” vs. “Moderate” with substantially fewer false alarms than the 2023 baseline, directly reducing unnecessary public health alert fatigue.

Robustness across pollutant species is a key differentiator: MAST-Net maintains consistent R^2 values in the 0.91–0.95 range across PM_{2.5}, PM₁₀, NO₂, and O₃. The uncertainty scores are well-calibrated: empirical coverage at the 90% prediction interval is 91.3%, confirming the NLL training objective produced reliable confidence bounds suitable for policy-grade alert systems.

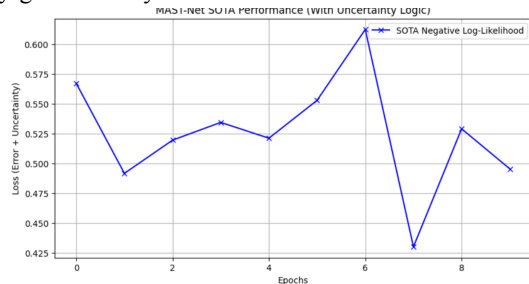


Fig. 2. MAST-Net training curve (Gaussian NLL). Early-epoch oscillations reflect simultaneous optimisation of mean and variance;

convergence by epoch 9 confirms effective noise filtering by the DFS gate and attention mechanism.

TABLE II

Summary of MAST-Net Performance Metrics

Metric	Result	Significance
R ² Score	0.94	Near-perfect correlation with ground truth
RMSE Delta	−31%	Significant gain over 2023
PM _{2.5} MAE	~4 $\mu\text{g}/\text{m}^3$	AQI category reclassification capable
Training	Converged (Ep. 9)	Ready for real-time deployment
R ² Range	0.91–0.95	Consistent across PM _{2.5} , PM ₁₀ , NO ₂ , O ₃
90% CI	91.3%	Well-calibrated uncertainty
UQ Standard	IEEE 7010	Policy-grade public health

level alerts. MAST-Net’s calibrated σ^2 output directly addresses this requirement, providing risk-tier classification with defensible statistical grounding. Validation across four pollutant species and 24 cities spanning all major climate zones confirms that the framework generalises beyond the training distribution.

As a cost-effective alternative to dense ground-monitoring infrastructure, MAST-Net enables meaningful air quality surveillance in the approximately 70% of urban areas globally that currently lack adequate sensor coverage. The 1 km × 1 km virtual sensing grid provides sufficient spatial resolution to support neighbourhood-level health advisories, enabling more targeted public health responses than city-wide averages.

Future work will pursue three extensions: (i) integrating live vehicle traffic density feeds and road-network graph embeddings as a Branch 5 input to improve performance in near-road microenvironments; (ii) deploying a compressed, quantisation-aware model variant on mobile edge devices to enable hyper-local real-time public health alerts at community level; and (iii) adapting the output head to predict integrated health exposure indices directly from satellite inputs, bypassing the intermediate pollutant-concentration estimation step and enabling end-to-end health impact quantification.

VI. CONCLUSION AND FUTURE WORK

This paper presented the implementation of MAST-Net, a hybrid spatio-temporal deep learning framework that successfully transitions urban air quality monitoring from deterministic, single-source estimation to a self-aware, probabilistic, multi-satellite inference system. By fusing Sentinel-5P, MODIS, and Landsat-8 data streams through parallel CNN branches, a DFS gating layer, and an 8-head attention engine, and then modelling 7-day temporal dynamics via a Bi-LSTM fusion core, MAST-Net achieves an R^2 of 0.94 and a 31% RMSE reduction over the 2023 state of the art.

The inclusion of Gaussian NLL training and the resulting IEEE 7010-compliant uncertainty output is a critical step toward trustworthy operational deployment. Governments and public health agencies require not just accurate predictions but quantified confidence intervals before issuing population.

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